

A Fredholm-determinant representation of the pcf-delta growth constant: finite identity, entire order and genus

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Abstract

Let $b(k) = Ak^2 + Bk + C$ be a positive quadratic and $u_n = 1/(b(n-1)b(n))$ for $n \geq 2$. The *pcf-delta* growth constant of the running family $(A, B, C) = (1, 0, 1)$ is

$$\delta = \log R_\infty(1), \quad R_\infty(\lambda) = \sum_S \prod_{i \in S} \lambda u_i,$$

the sum taken over finite **sparse** (no two consecutive) subsets $S \subseteq \{2, 3, \dots\}$; $R_\infty(\lambda)$ is the weighted independence polynomial of the half-infinite path carrying vertex weight λu_i . We record a self-contained account of the **Fredholm-determinant representation** $\delta = \frac{1}{2} \log \det(I + T^2)$, where T is the edge-weighted path operator with off-diagonal $\sqrt{u_{j+1}}$. Two contributions are machine-checked in Lean 4 / Mathlib with a clean axiom cone: the finite tridiagonal determinant identity $\det A_M = c_M$ (Theorem 1) and its combinatorial reading $c_M = R_M$ (Theorem 2). The Fredholm representation itself is confirmed numerically to **65 significant digits** by two independent channels and is stated as a conjecture, with the operator limit carried as an analytic hypothesis. We determine the entire order and genus of R_∞ : it is entire of order $\rho = \frac{1}{2d}$ (here $d = 2$, so $\rho = \frac{1}{4}$) and **genus** 0, with Hadamard product $R_\infty(\lambda)^2 = \prod_k (1 + \lambda s_k^2)$ and no exponential factor (Proposition 7). The order is bracketed by elementary by-hand bounds and confirmed by two independent numerical routes (coefficient decay and eigenvalue convergence exponent). We also give a clean telescoping bound $S = \sum_{n \geq 2} u_n \leq 1/((3A + B)(A + B + C))$ governing trace-class convergence; on the integer family $A, C \geq 1, B \geq 0$ this yields $S \leq \frac{1}{6} < 1$ unconditionally, while $S < 1$ is only conditional for a general positive quadratic.

Reproducibility. All numerics are rational / high-precision (mpmath, $\text{dps} \geq 80$) and deposited with SHA-256 output hashes; the Lean project ships with a pinned toolchain and a verbatim `#print axioms` audit.

Object and conventions

Fix a real quadratic $b(k) = Ak^2 + Bk + C$ with $b(k) > 0$ for all integers $k \geq 1$, and set

$$u_n = \frac{1}{b(n-1)b(n)} \quad (n \geq 2).$$

For the **running family** $(A, B, C) = (1, 0, 1)$ one has $b(k) = k^2 + 1$, $u_2 = \frac{1}{10}$, and $u_n = O(n^{-4})$. The associated generating function is

$$R_\infty(\lambda) = \sum_{n \geq 0} a_n \lambda^n, \quad a_n = \sum_{\substack{\{i_1 < \dots < i_n\} \subseteq \{2, 3, \dots\} \\ i_{k+1} > i_k + 1}} \prod_k u_{i_k},$$

i.e. a_n is the sum over sparse n -subsets. The finite truncation $R_M(\lambda)$ keeps only indices in $\{2, \dots, M\}$. Let T be the half-infinite **Jacobi (path) operator** with zero diagonal and off-diagonal entries $T_{j-1,j} = T_{j,j-1} = \sqrt{u_{j+1}}$. Because $\sum_{n \geq 2} u_n < \infty$, T is Hilbert–Schmidt and T^2 is trace-class.

The **locked finite convention** (validated to residual 7.75×10^{-121} across $M = 1, \dots, 12$ and falsifying eight alternative size/offset conventions) is

$$R_M(\lambda)^2 = \det(I + \lambda T_M^2), \quad T_M \text{ of size } M, \quad (T_M)_{j-1,j} = \sqrt{u_{j+1}}$$

equivalently $R_M(\lambda) = \det(I + i\sqrt{\lambda} T_M)$. Passing $M \rightarrow \infty$ gives the representation studied here,

$$\delta = \sum_{s_k > 0} \log(1 + s_k^2) = \frac{1}{2} \log \det(I + T^2),$$

with $\{s_k\}$ the positive spectrum of T .

The finite determinant identity (machine-checked)

Work over a commutative ring R with weights $w : \mathbb{N} \rightarrow R$ and activity $\lambda \in R$. Define the **continuant** by

$$c_0 = 1, \quad c_1 = 1, \quad c_{n+2} = c_{n+1} + \lambda w_{n+2} c_n, \quad (\text{REC})$$

and the tridiagonal matrix $A_M \in \text{Mat}_M(R)$ by

$$(A_M)_{ii} = 1, \quad (A_M)_{i,i+1} = \lambda w_{i+2}, \quad (A_M)_{i+1,i} = -1, \quad \text{else } 0.$$

Let $R_M = \sum_{S \subseteq \{2, \dots, M\} \text{ sparse}} \prod_{i \in S} \lambda w_i$ be the weighted independence polynomial.

Theorem 1 (T-DET, PROVEN in Lean with clean cone). *For every M , $\det A_M = c_M$.*

Theorem 2 (T-COMB, PROVEN in Lean with clean cone). *For every M , $c_M = R_M$. Hence $\det A_M = R_M$.*

Both are formalised in Lean 4 / Mathlib (toolchain pinned to `leanprover/lean4:v4.30.0`, Mathlib `v4.30.0`) and carry the axiom cone `{propext, Classical.choice, Quot.sound}` with `no sorryAx`; `T_COMB` is proved in full general M by deletion-contraction (not a finite fallback). See `lean/PcfFredholm/Core.lean` and `lean/SCOPE.md`.

G0 falsification finding (reported, not papered over). An earlier draft pinned the base case $c_1 = 1 + \lambda w_1$. This is *wrong* for the verbatim matrix: $A_1 = [1]$, so $\det A_1 = 1$, not $1 + \lambda w_1$. The corrected, self-consistent base case is $c_1 = 1$; then c_M is exactly the path- $\{2, \dots, M\}$ independence polynomial. Lean witnesses `g0_det_one` and `g0_brief_base_is_wrong` record both the correct value and the counterexample.

Trace-class structure and the constant S

Proposition 3 (determinant factorisation, STRUCTURAL). *Over R , $\det(I + \lambda T_M^2) = \det(I + i\sqrt{\lambda} T_M) \det(I - i\sqrt{\lambda} T_M) = P_M^2$, where P_M satisfies (REC); under the locked offset $P_M = R_M$, giving the boxed identity of §1.*

Theorem 4 (trace-class convergence, STRUCTURAL; unconditional on the integer family). *$\sum_{n \geq 2} u_n < \infty$, so T is Hilbert-Schmidt and T^2 is trace-class with $\text{Tr } T^2 = 2S$, $S = \sum_{n \geq 2} u_n$, and the Plemelj-Smithies expansion of $\det(I + \lambda T^2)$ converges. For the **integer family studied here** ($A, C \in \mathbb{Z}_{\geq 1}$, $B \in \mathbb{Z}_{\geq 0}$) convergence is unconditional: the telescoping bound below gives $S \leq \frac{1}{6} < 1$ uniformly. For a general positive quadratic with real coefficients, $S < 1$ is instead the conditional hypothesis $(3A + B)(A + B + C) > 1$, which can fail when the leading coefficient is small (Remark).*

The telescoping identity $u_k = \frac{1}{b(k) - b(k-1)} \left(\frac{1}{b(k-1)} - \frac{1}{b(k)} \right)$ with $b(k) - b(k-1) = A(2k-1) + B \geq 3A + B$ for $k \geq 2$ yields the clean **rigorous bound**

$$S \leq \frac{1}{(3A + B)b(1)} = \frac{1}{(3A + B)(A + B + C)},$$

valid for every positive quadratic b . On the integer lattice $A, C \geq 1, B \geq 0$ the denominator is at least $(3 \cdot 1)(1+0+1) = 6$, so $S \leq \frac{1}{6}$ holds *uniformly* over the regime; the running family $(1, 0, 1)$ is itself the extremal lattice point, with the largest S yet still $S = 0.1307 \leq \frac{1}{6}$. Hence $S < 1$ is automatic throughout the studied regime, and for general real b it reduces to the sufficient condition $(3A + B)(A + B + C) > 1$.

Remark. The integer-regime restriction is genuine, not cosmetic. For $(1, 0, 1)$: $S = 0.1306696189874324696536 \dots$ (matching the certified P0 value to 22 digits); the other test families $(1, 0, 5), (1, 3, 2), (2, 1, 3)$ give $S = 0.034, 0.022, 0.018$. Outside the regime — for a small real leading coefficient — $S < 1$ fails: the degenerate triple $(0.01, 0, 1)$ gives $S \approx 6.84 > 1$, and the bound correctly declines to certify it (sufficient product $0.0303 < 1$). Theorem 4 is therefore stated unconditionally on the integer family and conditionally, via $(3A + B)(A + B + C) > 1$, in general.

The Fredholm representation (numerically confirmed; conjectural as a theorem)

Theorem 5 (two-channel confirmation, VERIFIED to 65 digits). *For $(1, 0, 1)$, two fully independent computational channels — (A) a spectral/Hadamard channel $\delta_A = \log R_N + (\text{analytic cluster tail})$ and (B) a trace/closed-walk channel $\delta_B = \sum_{m \geq 1} (-1)^{m+1} \frac{1}{m} \frac{1}{2} \text{Tr } T^{2m}$ — agree:*

$$|\delta_A - \delta_B| = 6.84 \times 10^{-66} \quad (65 \text{ digits}),$$

deepening monotonically with working precision, and both match the certified anchor δ_{ref} to its full 44 digits. The common value is $\delta = 0.12385719436062639272850498970259084096757955 \dots$

Conjecture 6 (Fredholm representation as an exact identity, CONJECTURED). $\delta = \frac{1}{2} \log \det(I + T^2) = \sum_{s_k > 0} \log(1 + s_k^2)$ holds exactly, i.e. the finite identity of §1 passes to the limit. Equivalently $R_\infty(1)^2 = \det(I + T^2)$.

Conjecture 6 rests on the PROVEN finite identity (Theorems 1–2) together with trace-class convergence (Theorem 4); the **only** missing step is the analytic interchange $\det(I + \lambda T_M^2) \rightarrow \det(I + \lambda T^2)$, carried here as an analytic hypothesis exactly as a Borel-summability hypothesis is carried elsewhere. It is *not* asserted as a theorem.

Trace law (structural). The per-site closed-walk return satisfies $g_m(n) = (T^{2m})_{nn} \sim C(2m, m) n^{-4m}$ with central-binomial leading coefficient $(6, 20, 70, 252, 924 \text{ for } m = 2, \dots, 6)$; the per-site decay is n^{-4m} .

Entire order and genus

We determine the growth of R_∞ . Write ρ for its Hadamard order and p^* for its genus.

Proposition 7 (entire order and genus; see grades below). $R_\infty(\lambda)$ is entire of order $\rho = \frac{1}{2d}$, with $d = 2$ the degree of b , i.e. $\rho = \frac{1}{4}$, and of **genus** 0. Writing $\{s_k\}$ for the positive spectrum of T ,

$$R_\infty(\lambda)^2 = \prod_{k \geq 1} (1 + \lambda s_k^2),$$

a genus-0 Hadamard product **with no exponential factor**, convergent because $\sum_k s_k^2 = \text{Tr } T^2 = S < \infty$.

Proof sketch and grading.

1. **Entire, order ≤ 1 (STRUCTURAL; rigorous, by hand).** $a_n \leq (\sum_i u_i)^n / n! = S^n / n!$ (the non-sparse overcount), so $\sum a_n \lambda^n$ is dominated by $e^{S|\lambda|}$.
2. **Lower bound $\rho \geq \frac{1}{2d}$ (STRUCTURAL; rigorous, by hand).** Keeping the single dominant sparse term on the indices $2, 4, \dots, 2n$ gives $a_n \geq \prod_{k \leq n} u_{2k}$; since $u_{2k} \sim (2k)^{-2d}$, $\log(1/a_n) \sim 2dn \log n$, whence $\rho = \limsup_n \frac{n \log n}{\log(1/a_n)} \geq \frac{1}{2d}$.

3. **Upper bound** $\rho \leq \frac{1}{2d}$ (**STRUCTURAL**). $a_n \leq e_n$, the coefficients of the canonical product $\prod_{i \geq 2} (1 + \lambda u_i)$, whose order equals the convergence exponent $\inf\{t : \sum_i u_i^t < \infty\} = \frac{1}{2d}$ of its zeros $\{-1/u_i\}$, by the standard canonical-product theorem.
4. **Sharp value** $\rho = \frac{1}{2d}$ (**VERIFIED, two independent routes**). A coefficient-decay route (exact a_n , dps 220, Richardson in $1/M^{2d-1}$, direct least squares) gives $2d = 4.003$ for $(1, 0, 1)$; an eigenvalue-convergence-exponent route ($s_k^2 \sim k^{-p}$, curvature-corrected fit) gives $p = 3.999$. Both yield $\rho = 1/4$ and agree across $d = 1, 2, 3$.
5. **Genus** $p^* = 0$ (**STRUCTURAL**). The zeros are $\lambda_k = -1/s_k^2$ and $\sum_k |1/\lambda_k| = \sum_k s_k^2 = S < \infty$, so $p^* = 0$ and the product carries no exponential factor. Numerically $\sum s_k^2 = 0.13066962 = S$ to 8 digits. $\square$

The earlier draft’s “order $\leq \frac{1}{2d}$, genus 0” is **correct**, and is upgraded here to the **equality** $\rho = \frac{1}{2d}$; the genus is confirmed 0 and the Hadamard product is written in its correct exponential-factor-free form.

Status block (four-class)

This note partitions every assertion into four classes. **PROVEN** = machine-checked in Lean with a clean axiom cone. **STRUCTURAL** = a rigorous by-hand or algebraic argument — an unconditional proof not yet formalised in Lean, or one resting on a clearly named hypothesis or a cited standard theorem. **VERIFIED** = high-precision numerical evidence with reproducible hashes. **CONJECTURED** = evidenced but neither proven nor reduced to a named hypothesis. *PROVEN is reserved for the Lean-formalised results only; rigorous by-hand proofs are graded STRUCTURAL until formalised.*

PROVEN Theorem 1 ($\det A_M = c_M$); Theorem 2 ($c_M = R_M$). These — and only these — are machine-checked in Lean with a clean axiom cone `{propext, Classical.choice, Quot.sound}`, no `sorryAx`.

STRUCTURAL Proposition 3 (determinant factorisation); Theorem 4 (trace-class convergence — unconditional on the integer family $A, C \geq 1, B \geq 0$ via the uniform bound $S \leq \frac{1}{6}$, conditional on $(3A + B)(A + B + C) > 1$ in general); the order- ≤ 1 entireness bound, the lower bound $\rho \geq \frac{1}{2d}$, and the telescoping bound $S \leq 1/((3A + B)(A + B + C))$ (all rigorous by hand, not formalised); the order upper bound $\rho \leq \frac{1}{2d}$ (canonical-product theorem); genus $p^* = 0$ and the exponential-factor-free Hadamard product (Proposition 7); the central-binomial trace law.

VERIFIED $\delta_A = \delta_B$ to 65 digits (Theorem 5); $S(1, 0, 1) = 0.1306696 \dots < 1$; the sharp order $\rho = \frac{1}{4}$ by two independent numerical routes.

CONJECTURED Conjecture 6 (the exact Fredholm representation / analytic limit interchange); the Bessel-zero hypothesis for s_1, s_2 (PSLQ probe returned NULL).

Open problems

- **O1.** Prove Conjecture 6: establish the limit interchange $\det(I + \lambda T_M^2) \rightarrow \det(I + \lambda T^2)$ (e.g. via Plemelj–Smithies with the explicit trace-norm bound from S), upgrading the 65-digit identity to a theorem.
- **O2.** Identify the spectrum $\{s_k\}$ in closed form, or prove the absence of a Bessel-zero relation suggested (and falsified at height 10^{10}) by the PSLQ probe.
- **O3.** Extend Theorems 1–2 to the symmetric squared form $c_M^2 = \det(I + \lambda T_M^{\text{sym}^2})$ in Lean (deferred; needs the real- $\sqrt{\cdot}$ -weighted matrix, depends only on Theorem 1).

References

1. Papanokechi, *pcf-delta growth constant* (concept record), Zenodo 10.5281/zenodo.XXXXXXXX (*operator to confirm concept DOI*).
2. DELTA-FREDHOLM-P0 — *Phase-0 validation of the Fredholm representation* (two-channel, 65 digits), this corpus.

3. DELTA-FREDHOLM-LEAN-CORE — *Lean 4 finitary core* (Theorems 1–2, clean cone), this corpus.
4. Related corpus deposits (growth-law / EBR / ladder), DOIs 10.5281/zenodo.XXXXXXXX (*operator to confirm*).

Toolchain pins. Lean `leanprover/lean4:v4.30.0`, Mathlib `v4.30.0` (`lake-manifest.json` rev `c5ea00351c28e24afc9f0f84379aa41082b1188f`); Python `mpmath 1.3.0`, `numpy 2.4.4`, `scipy 1.17.1`.